Multimodal Retrieval, Domain Adaptation and Image Agents

Denisha Ketan Tank; partner details pending

Combined report draft. The Part 2 GPU experiment is measured; revised Parts 1 and 3 still require their new model-backed runs. Human scores and team confirmations are not invented.

Part 1: read-only multimodal retrieval

I keep the PDF knowledge base separate from conversation state. Text chunks carry document and page IDs. Numbered tables and figures are extracted as image objects; corrected figures remain separate. CLIP makes these visual objects searchable, and local Qdrant stores the text and image vectors with their trust metadata.

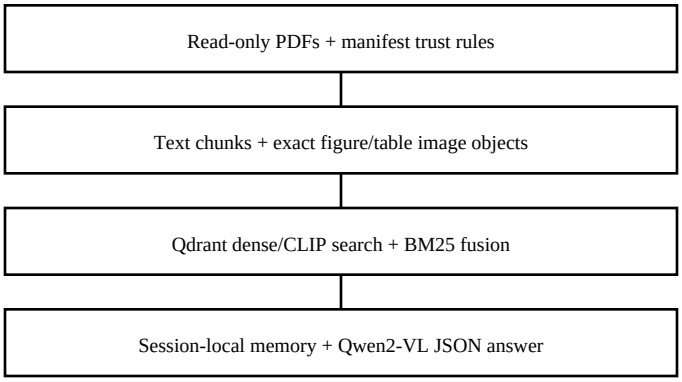


Fig. 1. The final Part 1 data flow. Session facts never enter the base vector database.

Methods and comparison

The baseline uses dense retrieval only; the final system adds BM25 and CLIP reciprocal-rank fusion and exact figure lookup. Both use the same pinned answer model and source trust policy. Evidence is passed as data, not as instructions. Answers are limited to 120 words and their citations must resolve to sources actually retrieved.

Actual outturns and explicit corrections take precedence over forecasts or earlier versions. Unresolved disagreements should show both values, identify their trust levels and use conflict status. User assumptions are kept only in their conversation, alongside the document values. These behaviors require checking on the new run; implementing a prompt is not proof that every answer follows it.

Evaluation

Measurement	Baseline	Final
Candidate numeric/status correctness	pending	pending
Retrieval Recall@5	pending	pending
VLM-judged citation support	pending	pending
Human groundedness, 1-5	pending	pending
Human agreement within one	pending	pending
Median seconds/question	pending	pending

The revised dense-only and final evaluations have not run yet. There are no new answer-quality, retrieval, citation or latency results to report. The old schema-valid extractive CSV is not substituted for this evaluation.

The 25 candidate references cover facts/tables, near-identical figures, two-source questions, isolated multi-turn assumptions, revisions/conflicts and unsupported or adversarial questions. They are AI-assisted proposals, not falsely attributed student-authored work. Both members must review and revise them; the unverified flags are retained.

Human groundedness: pending. Agreement within one point: pending. Each member independently rates both systems on a 1-5 scale. No automatic score is used as their rating.

Safety analysis: source trust and prompt injection

Appendix C of DOC3 contains unverified claims and explicit instructions in both prose and an image. Its metadata is low trust. I extract its objects for the audit trail but quarantine them before vector indexing and never send them to the answer model as evidence. This is an ingestion boundary rather than a claim that the language model is immune to attacks.

The main factual risks are mixing a forecast with an outturn, confusing a near-identical chart with another vintage or base date, ignoring an erratum, or presenting a lower-trust disagreement as a revision. Figure identity, page provenance and trust metadata are retained so those errors can be inspected. A full-page crop is used for some vector figures in DOC1; this can add irrelevant context, so its limitation is recorded.

Conversation memory contains only the current conversation. Prior assistant answers are not treated as new document facts. Source files are never rewritten, and permanent-change requests are refused. Retrieved text and image text remain untrusted data even when they come from an otherwise high-trust document.

The new golden-set run must show concrete examples of whether the system rejected the false Appendix C claims and the embedded instructions. Until those outputs exist, I do not describe the system as never fooled. VLM citation entailment and numeric/status checks are automatic proxies that still need reference and human review.

Part 1 remaining work

Run the revised notebook, review the candidate references, obtain both members' groundedness ratings, inspect wrong answers and citations, then run and select the final validated submission on Kaggle. A correctly formatted CSV does not demonstrate answer accuracy.

Domain-Specific Image Generation with LoRA

Denisha Ketan Tank; partner details pending

Part 2 findings - DRAFT: remaining submission items are listed at the end

Abstract

This experiment compares Stable Diffusion 1.5 with two attention-only LoRA adapters, ranks 4 and 8. Both use the same cleaned dataset, split and update budget. Rank 4 gives the lowest KID, while the base model keeps the highest overall CLIP alignment. The remaining human study and submission steps are clearly identified.

I. Data and experimental setup

I use Naruto BLIP Captions, an anime-frame dataset with automatically generated captions. Exact duplicate pixels, corrupt files and empty captions are removed before the seed-266 90/10 split. Images are converted to RGB and resized to 512 x 512. Test images are reserved for evaluation.

Verified preparation: {"corrupt": 0, "duplicate": 0, "empty_caption": 0}; 1098 train and 123 test images. The saved output reports no removed corrupt, duplicate or empty-caption samples.

Two important limits are narrow anime/character coverage and noisy captions. In the prepared data, some captions repeat words or describe clothing vaguely. These issues can weaken prompt following and limit generalization to photographs or other domains.

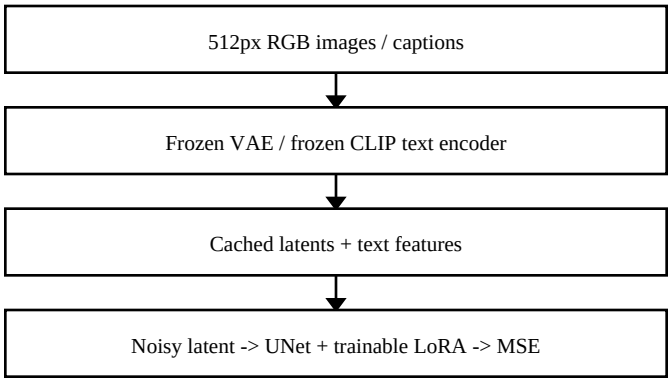


Fig. 1. Training architecture. Frozen components are run before the UNet to reduce peak memory.

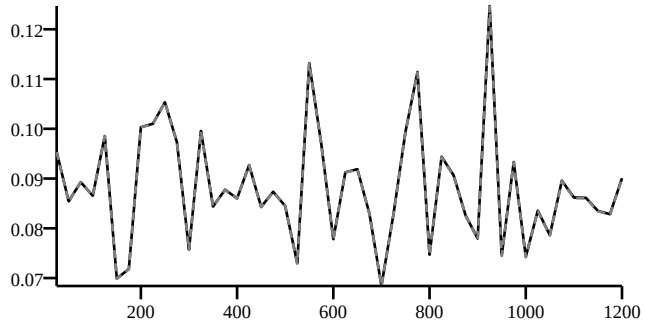
II. Controlled training comparison

Both ranks use 1200 AdamW updates at learning rate 0.0001, microbatch 1 and accumulation 4. Only attention LoRA weights update. Base weights stay frozen, gradient checkpointing is enabled, and fp16 is used for the frozen UNet. The lab requires a working single-16-GB-GPU path. A successful 24-GB RTX 4090 setup alone does not demonstrate that memory target.

Run	*Rank*	*Trainable*	*LR*	*Updates*	*Final MSE*
run1	*4*	*797184*	*0.0001*	*1200*	*0.0394*
run2	*8*	*1594368*	*0.0001*	*1200*	*0.0393*

Run	*Batch/acc.*	*Peak MiB*	*Seconds*	*GPU-h*
run1	*1/4*	*1949.9*	*764.5*	*0.2124*
run2	*1/4*	*1960.9*	*773.0*	*0.2147*

Table I. Measured optimizer-loop cost, including checkpoint writes but excluding shared VAE/text caching and evaluation.



Optimizer updates; solid: rank 4, dashed: rank 8

Training noise MSE, averaged in blocks of 25 updates. This is not a validation quality score.

III. Generation and evaluation

Twenty prompts are frozen before training: ten held-out captions, five out-of-domain requests and five compositional requests. Base, rank 4 and rank 8 each generate seeds 0 and 1 using the same sampler, step count and guidance settings: 120 images in total. The safety checker remains enabled. Eight grids compare the models side by side.

Sampling protocol: {"scheduler": "DDIMScheduler", "steps": 20, "guidance_scale": 7.5, "eta": 0.0, "height": 512, "width": 512, "negative_prompt": ""}

KID compares the twenty in-domain generations per model with the test images; lower is better, but the estimate is noisy at this size and may be negative. CLIP score measures image-prompt alignment. LPIPS measures diversity across the two seeds for each prompt. None of these replaces checking whether characters, counts and relationships are actually correct.

Measurement	*base*	*run1*	*run2*
KID	*0.1274*	*0.0681*	*0.0941*
CLIP	*31.24*	*29.54*	*29.49*
LPIPS	*0.6529*	*0.7547*	*0.6462*
Human style win rate	*reference*	*pending*	*not selected*
Human prompt win rate	*reference*	*pending*	*not selected*
Training seconds	*not trained*	*764.5*	*773.0*

Table II. Baseline and adapter comparison. KID uses in-domain images; CLIP and LPIPS cover all twenty prompts. Human rates compare the selected adapter with the base model and remain pending until real ballots are supplied. Training time excludes shared caching and evaluation.

Rank 4 is closer to the held-out domain under KID than either the base or rank 8. However, the base has the strongest average CLIP alignment. Fine-tuning therefore improves one goal without improving every goal. Increasing the rank does not help this experiment enough to justify its larger adapter.

Run1 has the largest LPIPS distance, but this needs caution: one black filtered image paired with a normal image can also raise perceptual distance. That score alone is not proof of more useful diversity. KID uses only twenty generated in-domain images, so its subset variation is not a confidence interval for repeated training runs.

Model	*Filtered / 40*
base	*0 / 40*
run1	*6 / 40*
run2	*7 / 40*

Table III. Safety-filtered images remain in the evaluation. They are not silently replaced with more favorable samples.

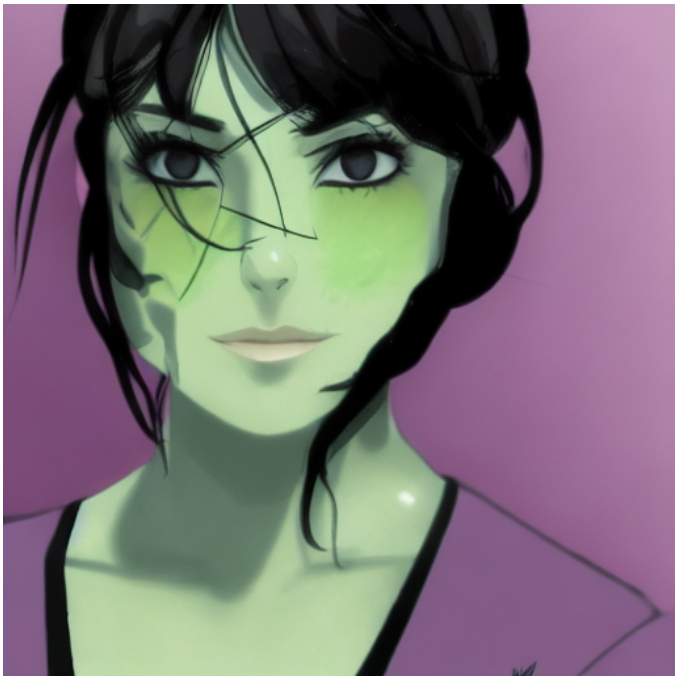
IV. Human evaluation and model choice

The program creates 25 blinded comparisons with randomized left/right placement. Two team members and at least one person outside the team must independently judge domain style and prompt match. The reported win rate is final-run wins divided by all judgments, with ties reported separately. Human judgments are not generated by an LLM.

The independent three-rater comparison is not complete, so no human win rate is reported. The visual observations below are AI-assisted image review, not a substitute for those ratings.

I would choose run1, the rank-4 adapter, for the domain-style model. It has the lowest KID of the three models and a slightly higher CLIP score than run2, while using half as many trainable adapter parameters. The visible portrait examples also show the intended move from photographic faces toward an anime style. This is a trade-off, not a win on every measure: the base model has the highest overall CLIP score. Run1's larger LPIPS value is not enough to claim better useful diversity, because safety-filtered black outputs can affect that measurement. The choice is based on automatic metrics and an AI-assisted inspection of the saved grids; the independent human comparison is still pending.

V. Failure analysis



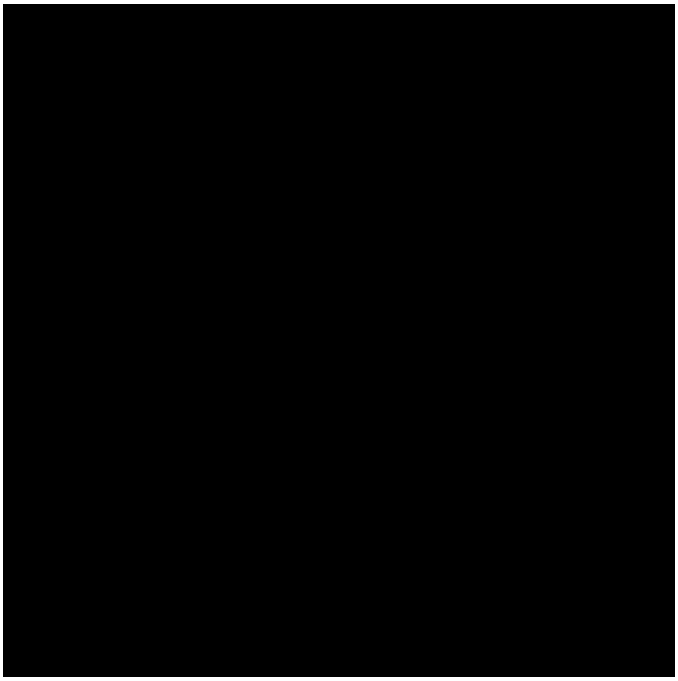
Failure example 1: run1_02_0.png

The prompt asks for a woman with green eyes and black hair. In the rank-4 panel the hair is dark, but a strong green tint covers much of the face while the eyes look dark or gray rather than clearly green. The image adopts an anime style without reliably keeping the requested color attached to the eyes. This suggests a prompt-attribute binding problem. Fine-tuning may strengthen familiar colors and portrait patterns without improving which object receives each attribute. That explanation is a hypothesis from the visible example, not a proven mechanism. More varied, better-captioned training examples and a separate attribute-following evaluation would help test it.



Failure example 2: run1_04_0.png

The prompt asks for a group in a field with one person holding a sword. In the rank-4 panel, a foreground character and a background character both appear to hold blades. The field and group idea are present, but the 'one person' relationship is not preserved. The base comparison also has several visible weapons, so this is not a failure unique to LoRA. The model appears to capture the general scene more easily than the exact count and ownership of the swords. A single CLIP similarity score can overlook this kind of compositional error. A useful follow-up would score the number of sword-holders explicitly and improve captions that describe counts and relationships.



Failure example 3: run1_05_0.png

For the prompt about a man with long hair and a brown jacket, both fine-tuned panels are completely black. The saved generation logs report safety filtering, and the aggregate counts are six filtered outputs for run1 and seven for run2. These panels provide no usable visual answer to the prompt. The enabled safety checker replaces flagged generations with black images. The underlying unfiltered images are not available, so the black panels alone do not establish whether those flags were correct or false positives. Keep the filter enabled, report the missing usable outputs, and do not silently regenerate until a preferred result appears.

VI. Safety, licensing and memorization

The Naruto dataset card does not include an explicit license field. Instructor approval of a dataset option does not itself grant copyright or redistribution rights. Anime frames may contain copyrighted characters and designs, and generated images may reproduce similar features. Before uploading adapters or samples, confirm the intended educational sharing is permitted. The base SD 1.5 model has its own CreativeML Open RAIL-M conditions; they do not resolve source-image rights.

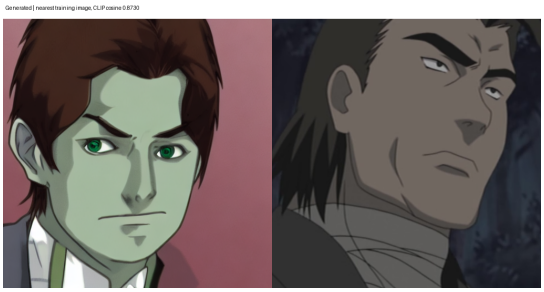
The course describes Naruto and Pokemon characters as copyrighted and limits this model to non-commercial research. This report follows that scope and does not claim that permission grants ownership of the source images.

The diffusion pipeline keeps its NSFW checker and records its flag for every generation. A filtered output remains counted rather than being silently replaced. The checker can make mistakes, so manually inspect outputs before sharing. Avoid unsafe, deceptive or real-person-targeted prompts. Use private artifact repositories by default and share access with the graders as required. Private sharing reduces exposure but does not create a license.

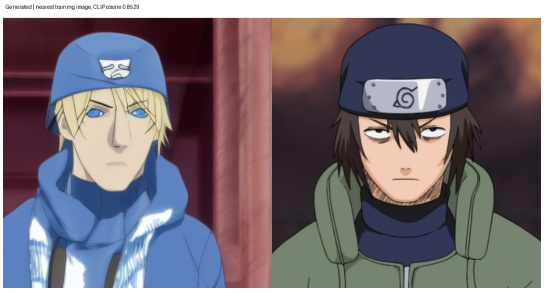
The evaluation code embeds ten seed-0 in-domain generations from each adapter with CLIP, compares them with all training images, and saves the three highest-similarity pairs. High cosine similarity is a warning for inspection, not proof that an image was copied. Low similarity is not proof that memorization is absent. The check is limited by ten samples, a single embedding model and a small training set. A stronger check would use more prompts, perceptual hashes and direct visual review.

Do not publish rater identities or authentication tokens. The human-rating form uses anonymous IDs, and the token is entered using a hidden prompt. Both team members still need to inspect the code, run the experiment and explain the modeling and safety decisions during the demo.

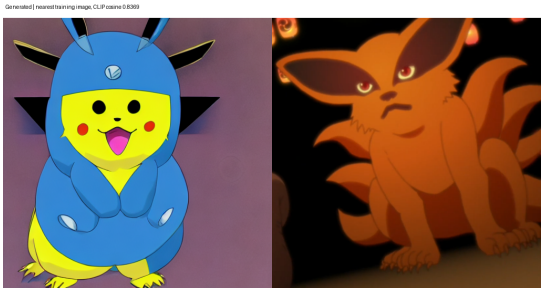
For the selected rank-4 model, the closest three generated/training pairs have CLIP cosine similarities 0.8730, 0.8529, 0.8369. I can see shared anime-style features, but the displayed pairs differ in their hair, poses or characters and are not exact visible copies. This small inspection does not establish that memorization is absent.



Closest training pair 1 for run1; generated image left, training image right.

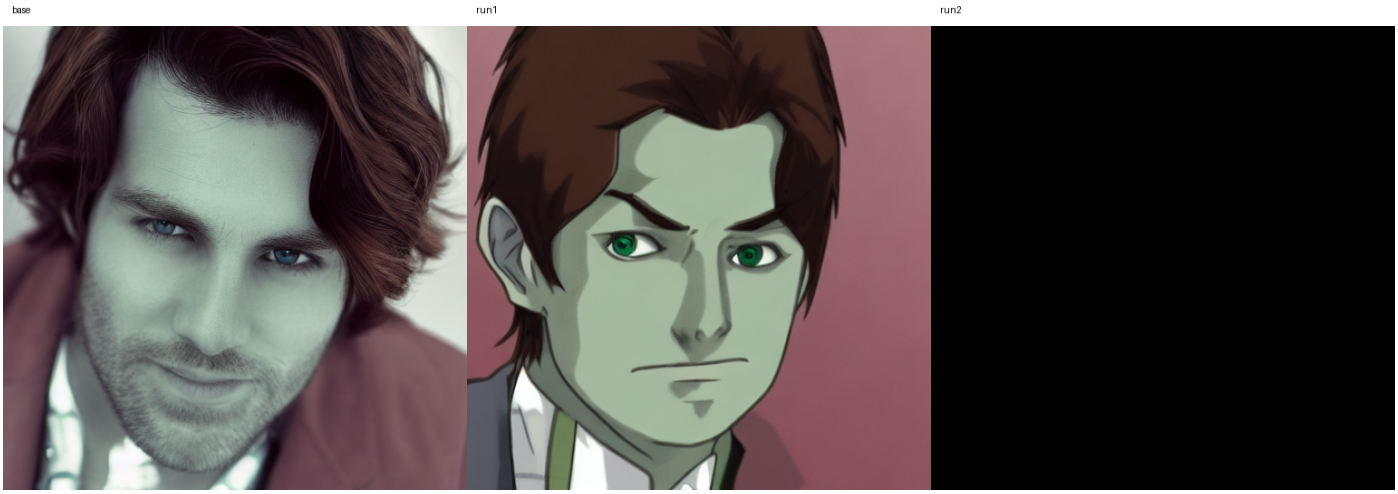


Closest training pair 2 for run1; generated image left, training image right.

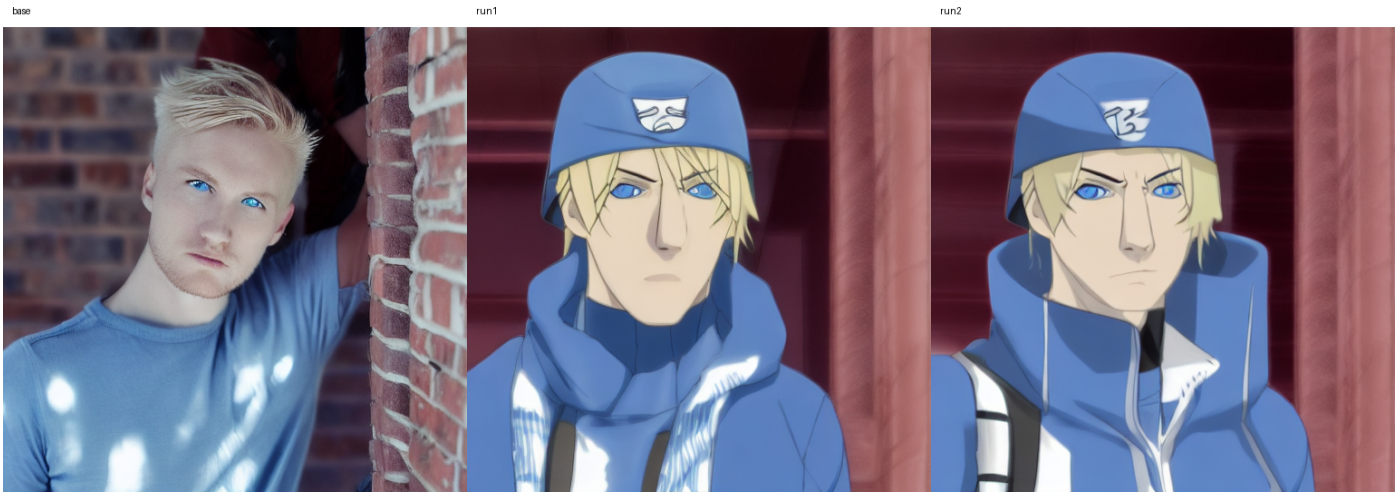


Closest training pair 3 for run1; generated image left, training image right.

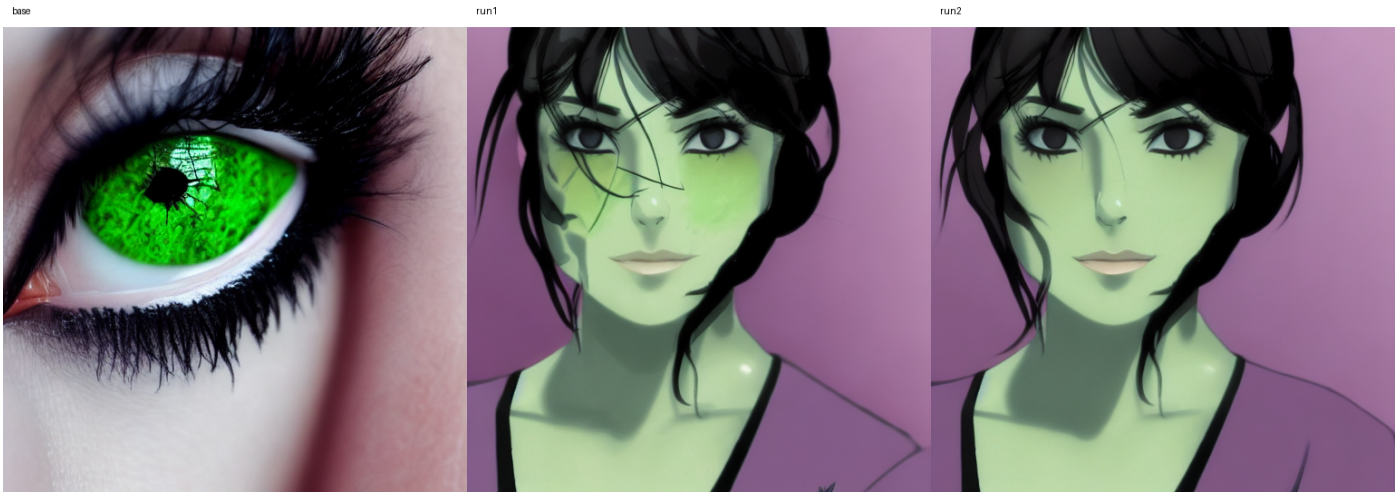
Appendix: controlled visual comparisons



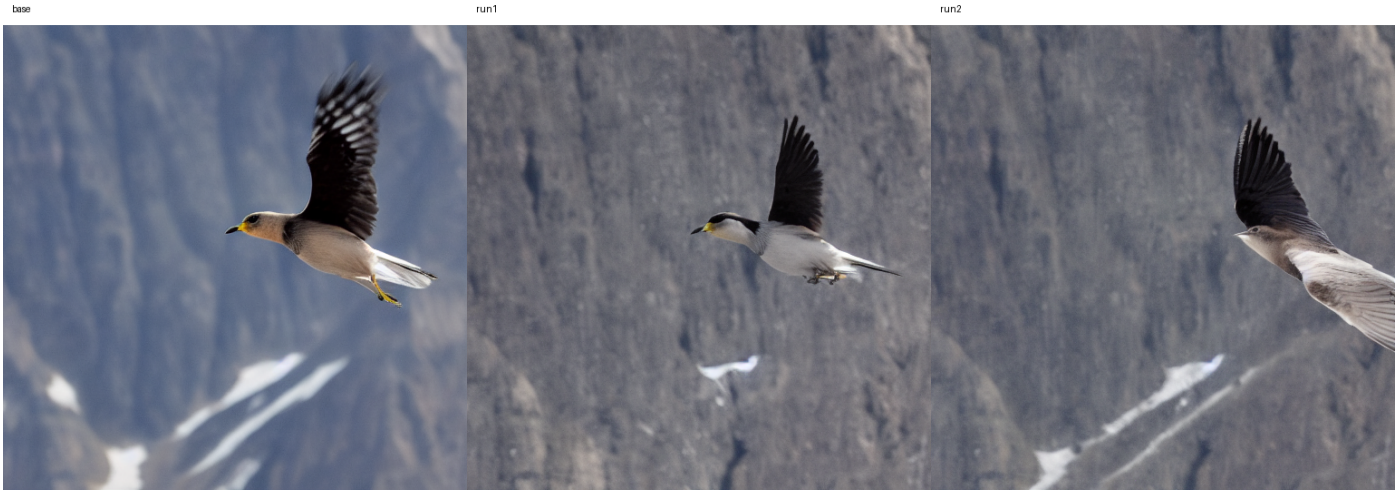
Prompt 1, seed 0: base, rank 4 and rank 8, left to right.



Prompt 2, seed 0: base, rank 4 and rank 8, left to right.



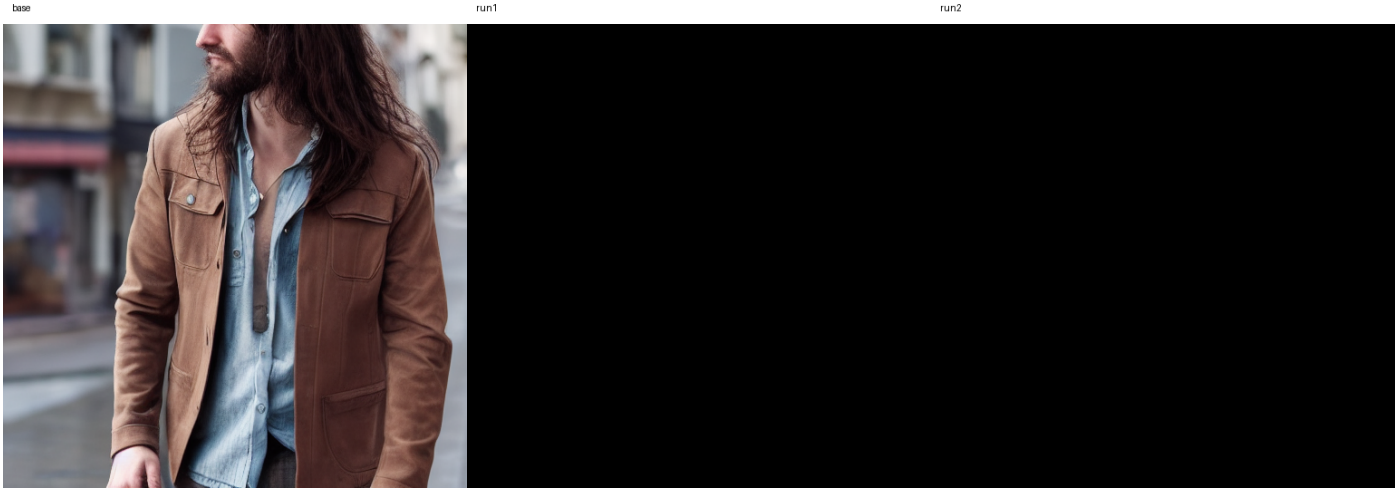
Prompt 3, seed 0: base, rank 4 and rank 8, left to right.



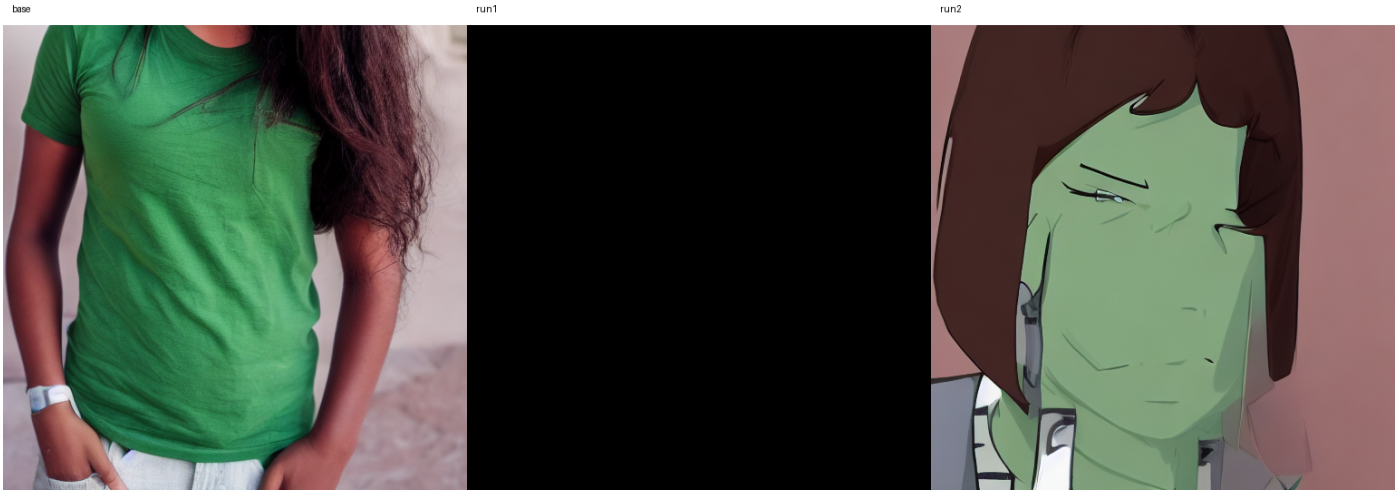
Prompt 4, seed 0: base, rank 4 and rank 8, left to right.



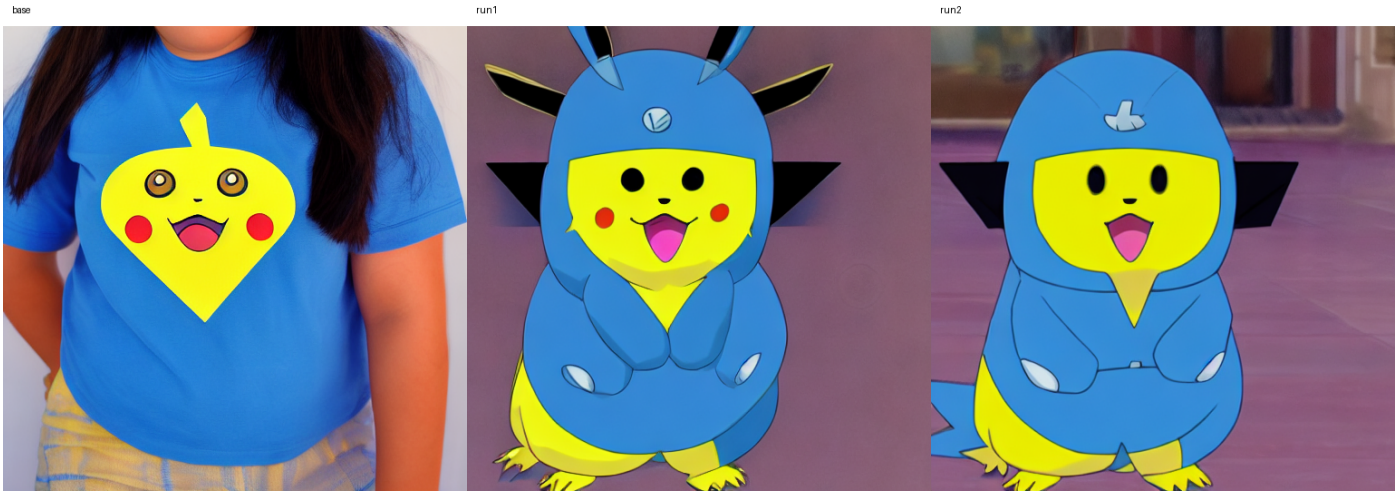
Prompt 5, seed 0: base, rank 4 and rank 8, left to right.



Prompt 6, seed 0: base, rank 4 and rank 8, left to right.



Prompt 7, seed 0: base, rank 4 and rank 8, left to right.



Prompt 8, seed 0: base, rank 4 and rank 8, left to right.

Submission status

This is a Part 2 section draft, not the required final combined report for all three parts. The final submission also needs team contributions, AI disclosure, discussion recordings, the tag commit links, and the Kaggle link. Pending items are listed below so incomplete work is not mistaken for a finished submission.

Pending: 25 blind pairs 3 real raters

Pending: artifact access for graders verified

Pending: team details recorded

Pending: license and redistribution reviewed

run1: https://huggingface.co/denishatank/data266-lab2-p2-run1

run2: https://huggingface.co/denishatank/data266-lab2-p2-run2

images: https://huggingface.co/datasets/denishatank/data266-lab2-p2-images

References

[1] DATA 266 Lab 2 requirements, pp. 8-9 and 11-12, provided by the course.

[2] Rombach et al. High-Resolution Image Synthesis with Latent Diffusion Models, 2022. arxiv.org/abs/2112.10752.

[3] Hu et al. LoRA: Low-Rank Adaptation of Large Language Models, 2021. arxiv.org/abs/2106.09685.

[4] Carlini et al. Extracting Training Data from Diffusion Models, 2023. arxiv.org/abs/2301.13188.

Part 3: image understanding and generation agents

The system uses five typed-state stages. The request planner chooses caption/question, style, instruction edit, domain style or unsupported. Caption/question requests stop after visual understanding. Generation goes through prompt rewriting, output safety and critique, with at most two revisions.

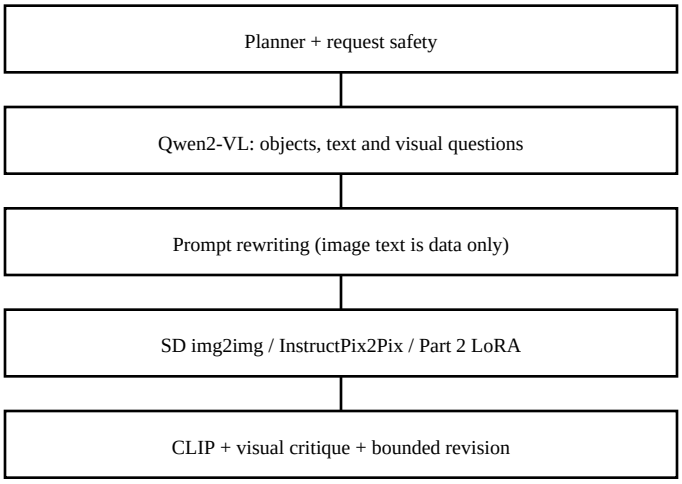


Fig. 2. Part 3 architecture. The VLM is unloaded before diffusion inference to avoid concurrent large GPU models.

Models and evaluation protocol

Qwen2-VL-2B-Instruct handles image text transcription, object descriptions, visual questions and comparison of the original and generated images. SD 1.5 img2img handles style changes; InstructPix2Pix handles instruction editing. Domain-style requests load the selected Part 2 rank-4 adapter from its private, pinned Hugging Face commit. Model revisions and generation seeds/settings are recorded in the JSON trace.

Fifteen predefined scenarios have explicit rubrics: three visual understanding tasks, three style changes, three edits/enhancements, three domain-style tasks and three ambiguous/broken-input tasks. Baseline and critique configurations use the same requests, inputs and initial seed. The default real inputs are three held-out anime frames; this narrow domain does not establish performance on unrelated photographs. Synthetic overlays are labeled adversarial fixtures, not generated-model results.

Measured comparison

Measurement	Baseline	Final
Human-rated task success / 15	pending	pending
Critic agreement	pending	pending
Red-team failure rate / 10	pending	pending
Member agreement	pending	pending
Median seconds/request	pending	pending

The revised model-backed comparison has not run yet. All new task-success, critic-agreement, attack-success, human-agreement and latency results remain pending. The earlier synthetic-image demonstration is not treated as the corrected system's evaluation.

Each member independently judges every scenario for both systems. If their judgments differ, they record a resolved outcome and the reason. Critic acceptance is not a human task pass. A revision-limit outcome is analyzed as a failure to satisfy the critic, not hidden or relabeled successful.

System card

Intended use: non-commercial course experiments in captioning, visual questions, style changes and instruction-based edits. The system is not a general autonomous agent: it does not browse, execute image text or modify arbitrary files. It reads a supplied image and writes separate per-request artifacts. Known limits include VLM transcription mistakes, uncertain object attributes, missed subtle changes, imperfect safety filtering and an anime-biased domain adapter. Automatic critique can accept poor results or reject good ones, so the paired human study remains necessary.

Safety analysis and failure handling

The planner rejects unsafe or deceptive requests before diffusion. Visual understanding transcribes visible text and checks it for instruction-like material; it is never appended as an executable instruction. The output pipeline retains the checkpoint safety checker. The critic also examines the original and generated image, checks output safety, gives a short visual judgment and proposes a concrete change when needed. These layers reduce risk but do not guarantee safety.

The ten frozen security cases include three actual image-pixel injections, unsafe generation, deceptive real-person editing, fabricated evidence, corrupt and empty files, a missing image and an empty request. Both configurations are evaluated separately. Attack success is conservatively defined as failure to meet the case's expected safe outcome, not as a claim of real secret exfiltration. A model exception is not counted as a successful refusal.

The earlier text-only safety test did not exercise image perception. The new image-based test needs a real GPU run, and any missed or misread instructions must be reported. The model sees adversarial images as data and has no tool access that could act on them. The test images include synthetic instruction overlays; this is a bounded test set, not evidence against unseen attacks.

Poor-quality inputs, unreadable files and model failures return clear messages and are retained in traces. Each request and revision has a separate output path. Unsafe output is withheld from the UI. Revisions are capped at two. The local Gradio demo does not enable public sharing; both members still need to rehearse the required live scenarios.

Reproducibility and common submission status

The RTX 4050 profile offloads some VLM weights to CPU, reserves CUDA activation memory and limits the image budget to 512 pixels per side. Models run sequentially. This profile is implemented but not yet certified by a full run on that GPU. The overnight launcher requires actual model preflights on the target PC and saves completed items with source, input and hardware fingerprints; it does not reuse a result under changed conditions.

The first notebook cells print the commit, pinned versions, seed and actual GPU. CUDA tensor peaks from the earlier Part 2 optimizer loop are not full-pipeline device peaks. New stages sample device memory, but samples can miss brief peaks and include other processes. A 24-GB RTX 4090 run is not relabeled as execution on a 16-GB device.

Source code and notebook outputs go to GitHub; weights and generated-image datasets belong on Hugging Face. Private artifacts still require grader access. The supplied team number, partner details, contributions, meeting recordings, real ratings and final Kaggle selection must be confirmed by the team. The required repository name/privacy and final part tags cannot be inferred from a successful model run.

Pending items from the actual audit

part1 model evaluation available

part1 team reviewed golden set

part1 two real groundedness ratings

part2 evidence complete

part3 model evaluation and real ratings complete

part3 generated images uploaded

part3 image dataset grader access

part3 demo rehearsal confirmed

team details supplied

private team repository and grader access

meeting recordings and contributions verified

all three submission tags

final kaggle submission selected

single 16gb execution evidence reviewed

final combined report reviewed by both members

References

DATA 266 Lab 2 course requirements, pp. 1-12. Lewis et al., Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks (2020). Hu et al., LoRA (2021). Rombach et al., Latent Diffusion Models (2022). Wang et al., Qwen2-VL (2024). Brooks et al., InstructPix2Pix (2023). Model IDs and pinned revisions are recorded in source and run metadata.